

Install WSL2 + Ubuntu + Ollama on Windows 11

```
Windows 11
  ↓
WSL2 Ubuntu
  ↓
Ollama
  ↓
Local AI Model
```

Final goal:

```
ollama run qwen2.5:3b
```

and ask:

```
hello
```

and receive an AI response locally on your machine.

Step 1 — Open PowerShell as Administrator

Search in Windows:

```
PowerShell
```

Right click → **Run as Administrator**

Step 2 — Install WSL2

Run:

```
wsl --install
```

This installs:

- Windows Subsystem for Linux
- WSL2
- Ubuntu

Windows may ask for restart. Restart your system if prompted.

Step 3 — Verify WSL2 Installation

After reboot, open PowerShell again and run:

```
wsl --status
```

Expected output:

```
Default Version: 2
```

This confirms WSL2 is installed correctly.

Step 4 — Open Ubuntu

Open Start Menu and search:

```
Ubuntu
```

Launch it.

The first launch may take a few minutes.

You will be asked to create:

- Linux username
- Linux password

Example:

```
Username: vishnu
```

Step 5 — Verify Ubuntu Running

Inside Ubuntu terminal run:

```
uname -a
```

Expected:

```
Linux ... microsoft-standard-WSL2
```

Ubuntu is now running inside Windows using WSL2.

Step 6 — Update Ubuntu

Run:

```
sudo apt update && sudo apt upgrade -y
```

Enter your Linux password if prompted.

Step 7 — Install Required Packages

```
sudo apt install -y curl git unzip build-essential
```

Step 8 — Install Ollama

Run:

```
curl -fsSL https://ollama.com/install.sh | sh
```

This installs:

- Ollama runtime
- Ollama server
- Ollama CLI

Step 9 — Verify Ollama Installation

Run:

```
ollama --version
```

Expected example:

```
ollama version is 0.24.0
```

Ollama installed successfully.

Step 10 — Start Ollama Server

Run:

```
ollama serve
```

Possible output:

```
Error: listen tcp 127.0.0.1:11434: bind: address already in use
```

This is actually okay. It means Ollama is already running in background.

Step 11 — Verify Ollama API Running

Open a NEW Ubuntu terminal window.

Run:

```
curl http://localhost:11434/api/tags
```

Expected output:

```
{"models":[]}
```

This confirms Ollama server is active.

Step 12 — Download Your First AI Model

Recommended beginner model:

```
ollama pull qwen2.5:3b
```

This downloads several GB. Wait until complete.

Depending on internet speed, this may take several minutes.

Step 13 — Verify Downloaded Models

Run:

```
ollama list
```

Expected:

NAME	SIZE
qwen2.5:3b	4.x GB

Step 14 — Run the AI Model

Run:

```
ollama run qwen2.5:3b
```

The model will start loading.

Now ask:

```
hello
```

Expected:

```
Hello! How can I help you today?
```

Congratulations — your local AI model is now working successfully.

Step 15 — Exit Model

Type:

```
/bye
```

Useful Ollama Commands

List Installed Models

```
ollama list
```

Run Model

```
ollama run qwen2.5:3b
```

Download Another Model

```
ollama pull llama3.1:8b
```

Remove Model

```
ollama rm qwen2.5:3b
```

Check Running Models

```
ollama ps
```

Recommended Models

Balanced Model

```
ollama pull qwen2.5:3b
```

Better Reasoning

```
ollama pull llama3.1:8b
```

Coding Model

```
ollama pull deepseek-coder:6.3b
```

Lightweight Fast Model

```
ollama pull phi3:mini
```

Recommended Hardware

- **Minimum:** 16 GB RAM
- **Recommended:** 32 GB RAM

Final Architecture

Windows 11



WSL2 Ubuntu



Ollama



Local AI Model

You now have a fully local AI setup running on your Windows machine.